



HARAMAYA UNIVERSITY
COLLEGE OF COMPUTING AND INFORMATICS
DEPARTMENT OF SOFTWARE ENGINEERING

SELECTED TOPICS IN SOFTWARE ENGINEERING (SENG 5112)

TOPIC: SCRUM METHODOLOGY

CHAPTER ONE- INTRODUCTION



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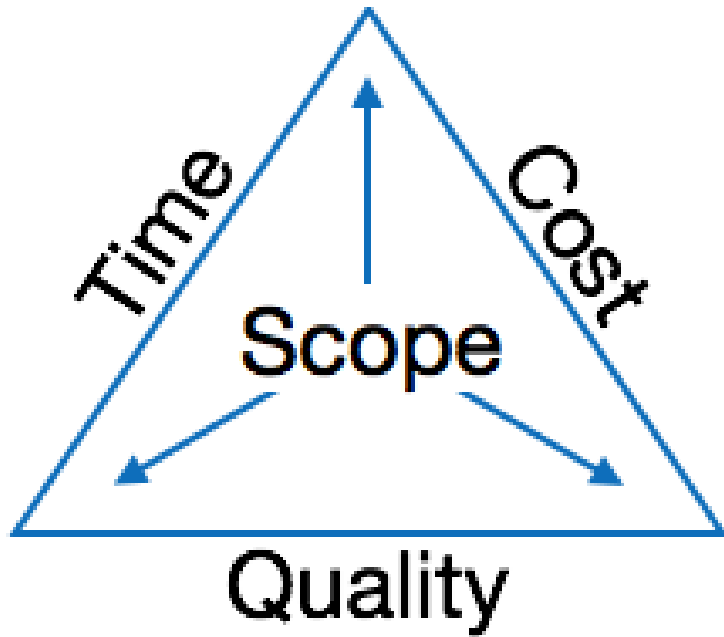


- ☐ Introduction
- ☐ What is Agile?
- ☐ What is Scrum?
- ☐ Scrum Theory
- ☐ Scrum Structure
- ☐ Scrum Values.
- ☐ Scrum Framework

INTRODUCTION ... (1)



- ❑ A software project involves the complete process of software development, from gathering requirements to testing and maintenance, carried out within a specified timeframe to achieve the desired product.
- ❑ Software is intangible, and its development is a relatively new field with limited experience in building software products.
- ❑ Most software products are custom-made to fit client requirements, and the rapid advancement of technology means that experience from one project may not apply to another.
- ❑ These factors introduce risks, making efficient software project management essential.



- ❑ It is an essential part of software organization to deliver **quality** product, keeping the **cost** within client's budget constrain and deliver the project as per **scheduled**.
- ❑ Therefore, **software project management** is essential to incorporate user requirements along with budget and time constraints.



- ❑ Delivering a quality product within the client's budget and timeline is crucial for software organizations.
- ❑ Software project management ensures that user requirements, budget constraints, and timelines are effectively incorporated.
- ❑ Project management methodologies provide principles, tools, and techniques to plan, execute, and manage projects, ensuring they are completed on time, within budget, and with high quality.
- ❑ Common methodologies include Agile (flexibility and speed) and the Waterfall model (linear and phase-dependent).

WHAT IS AGILE? ... (1)



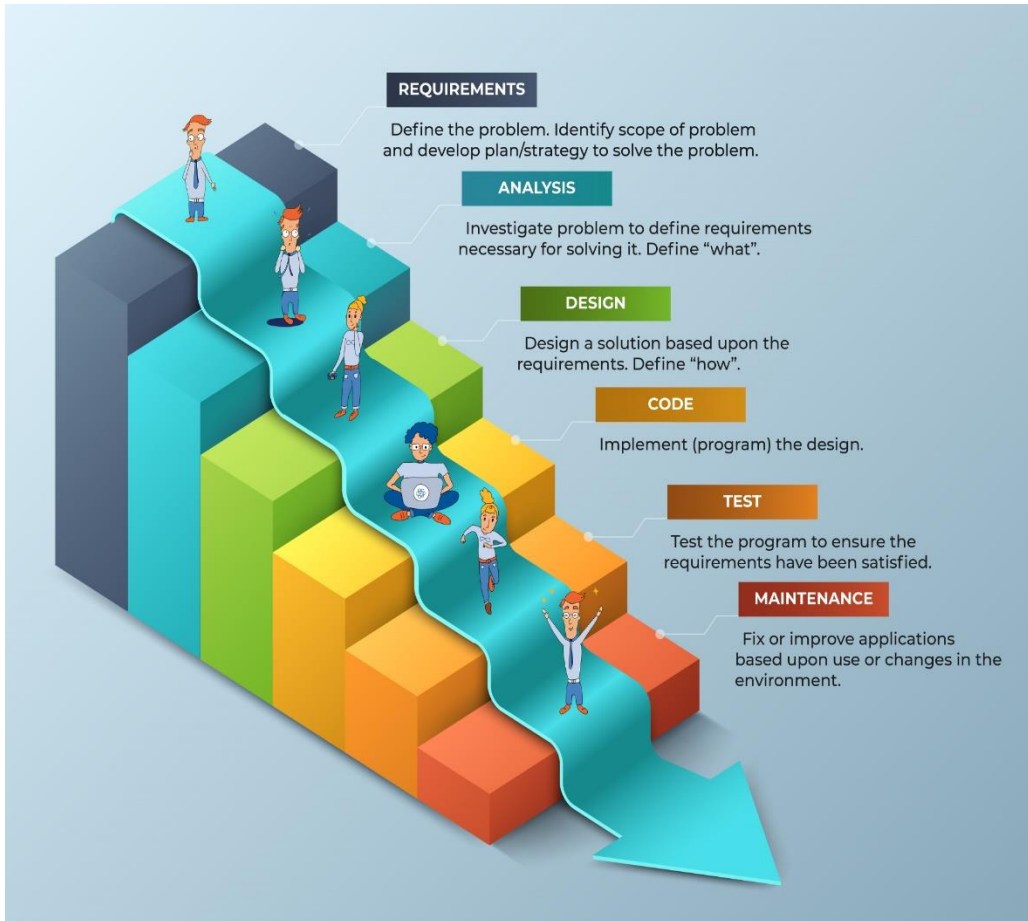
- ❑ Agile methodology is a **project management** approach that allows successful and efficient execution of the project while emphasizing the improvement of a project and team collaboration.
- ❑ The approach is applicable in software development for **flexibility, customer satisfaction, and collaboration.**
- ❑ It refers to the application of a set of principles that functions through an interactive and incremental approach.
- ❑ The Agile methodology in software development emphasizes the importance of team **collaboration** and delivering a working product **quickly** to meet customer needs and expectations.

WHAT IS AGILE? ... (2)

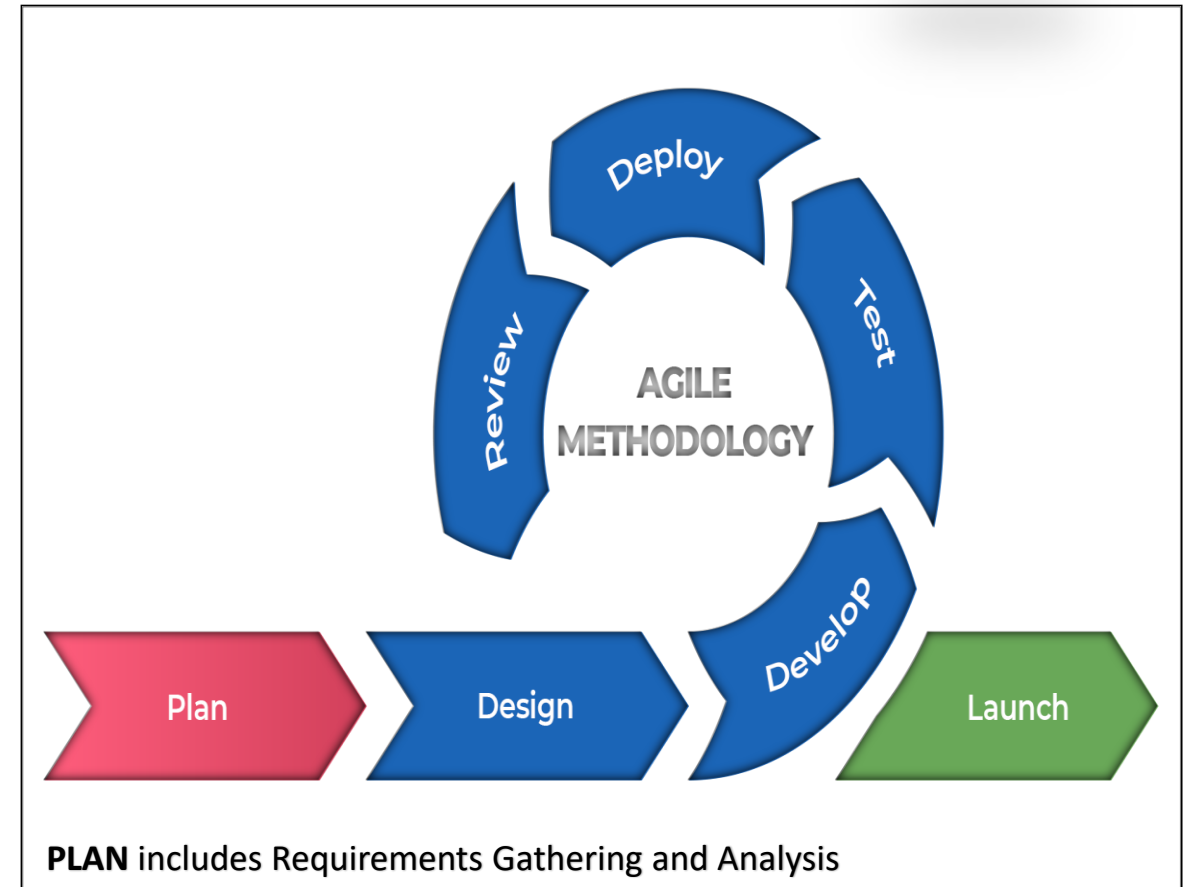


- ❑ Agile is a project management approach that emphasizes flexibility, customer satisfaction, and team collaboration.
- ❑ It focuses on iterative and incremental development, delivering working products in short cycles (iterations) rather than all at once.
- ❑ Agile is particularly suited for software development, where requirements and solutions evolve through collaboration between self-organizing, cross-functional teams.
- ❑ Popular Agile methodologies include **Scrum**, **Kanban**, **XP**, **Lean**, and **Crystal**.

WHAT IS AGILE? ... (3)



WATERFALL METHODOLOGY



AGILE METHODOLOGY



Core Principles of Agile Software Development Methodology

- ☐ Achieving **customer satisfaction** by timely and regular delivery fitting the customer requirements
- ☐ Delivering working software with **minimal** or **no errors**
- ☐ Earning a competitive edge by **adapting to the changes** in the market
- ☐ Sticking to the **shortest time delivery** with **quality**
- ☐ Including motivated and **passionate individuals** in building and developing the project
- ☐ **Accepting change** in requirements and deliveries despite nearness to delivery date
- ☐ Choosing the **face-to-face communication** method for efficient and effective information exchange.
- ☐ Comprehending excellence while estimating and improving the progress

WHAT IS SCRUM? ... (1)



Jeff Sutherland

Ken Schwaber

- ❑ Scrum is the most popular, modern, and flexible **agile framework** that enables teams to work together.
- ❑ Based on agile principles, scrum enables the development, delivery, and sustenance of complex projects.
- ❑ It enables teams to hypothesize how they think something works, try it out, learn and reflect from their experiences, and make appropriate changes.
- ❑ Its ever-growing popularity makes **scrum** the most proven organizational framework adapted by teams striving for maximum effectiveness.

WHAT IS SCRUM? ... (2)



Scrum is simple by design.

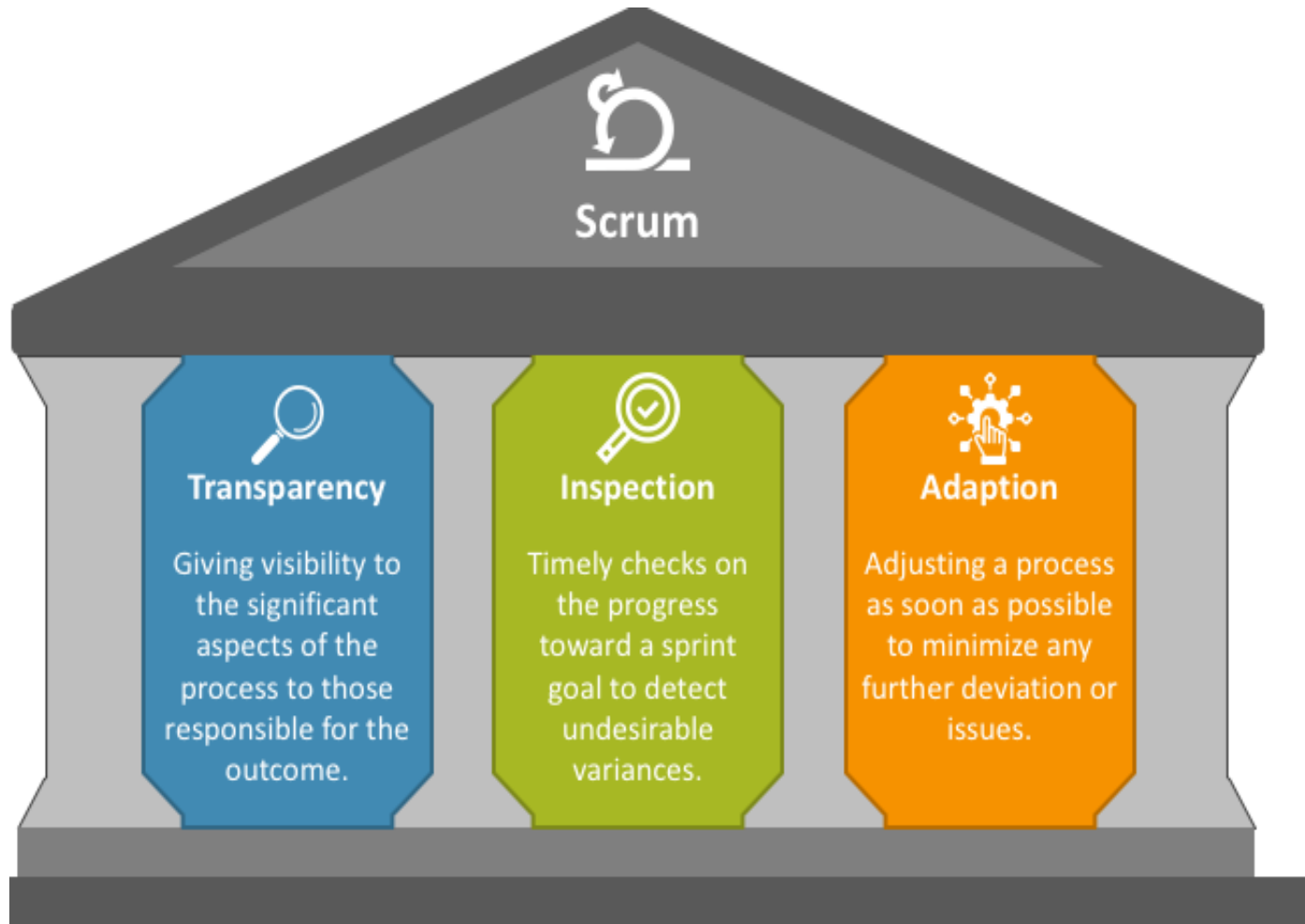
- ❑ **First**, it facilitates **breaking down** difficult problems into smaller, more understandable components.
- ❑ **Secondly**, it provides solid structure by dividing the work into stages to execute in given time frames, bringing **measurable** and **satisfying** results.
- ❑ **Thirdly**, it enables planning of the next stages of work using obtained results and conclusions drawn from the ongoing processes.
 - ❑ Scrum is comprehensive and serves to optimize teamwork.
 - ❑ It offers a clearly-defined starting point.
 - ❑ **The scrum philosophy focuses on the need for continuous development and reshaping through feedback, reflection and experience.**
 - ❑ It rejects complex, rigid systems that organize work without taking into account specific realities.



- ❑ Scrum theory's main principle concerns **empiricism**.
- ❑ **Empiricism** means keeping planning and assumptions to a minimum in favor of relying on experience, observation, and experimentation.
- ❑ It becomes possible and effective thanks to the iterative approach, i.e., working in short cycles, which includes not only working on the product, but also planning it and evaluating the results.

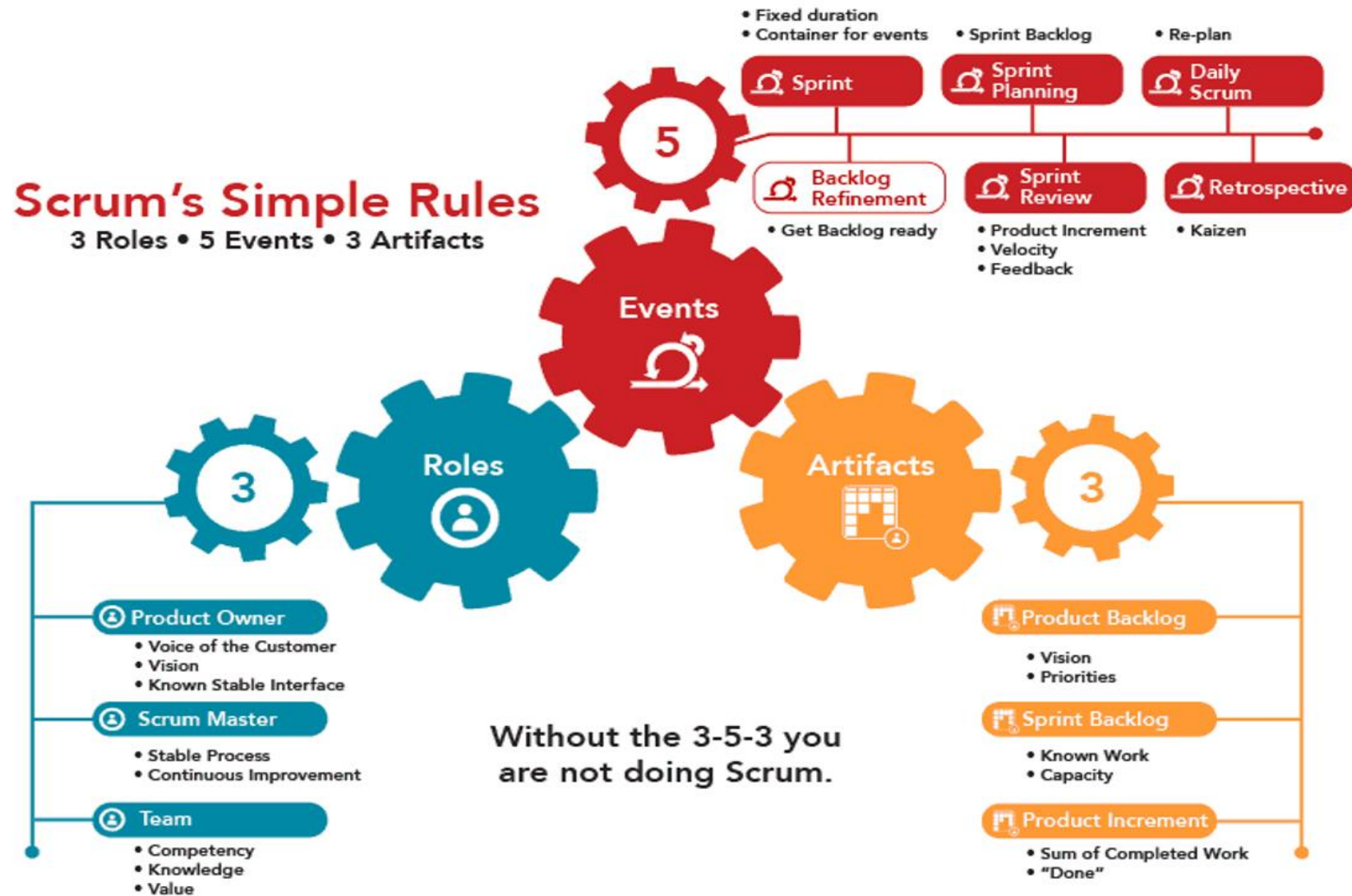


Three pillars of empiricism are the most important for Scrum's effectiveness: .

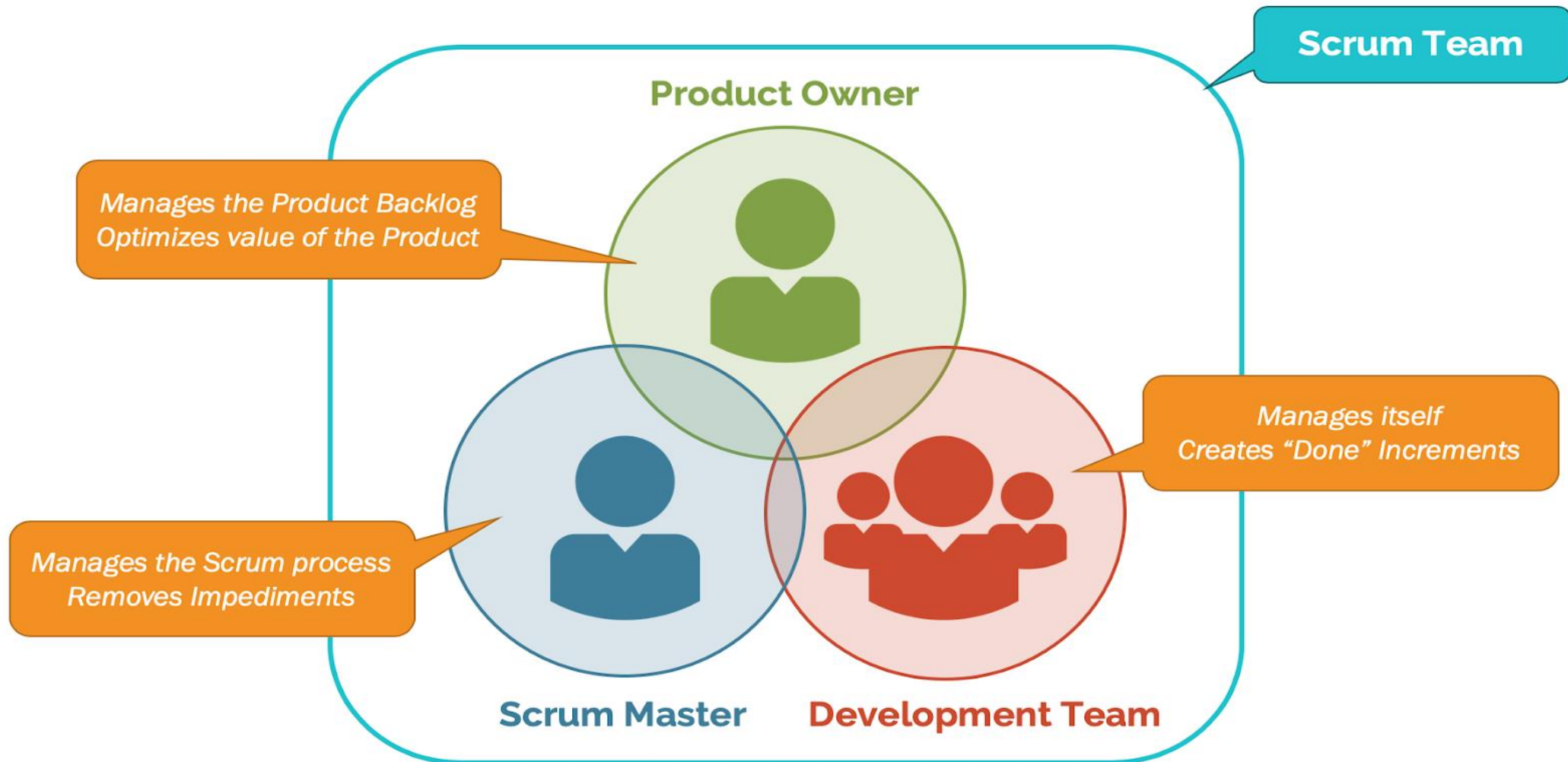


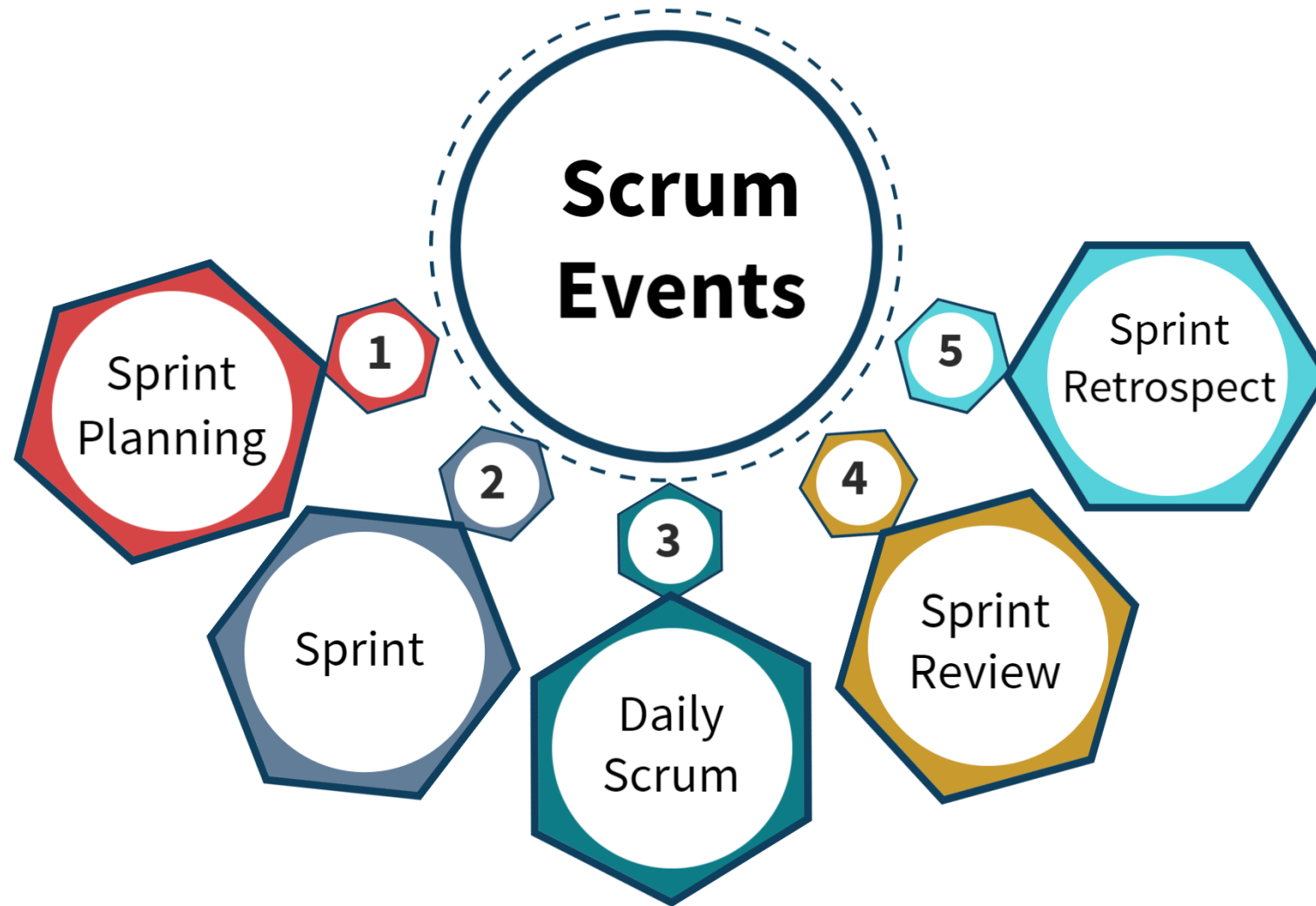
Empiricism works best if the team operating according to its principles has the ability to self-manage according to the lean concept. It implies a flexible organizational structure that allows **adaptation** to exist conditions, **continuous improvement**.

SCRUM STRUCTURE ... (1)



SCRUM STRUCTURE ... (2)





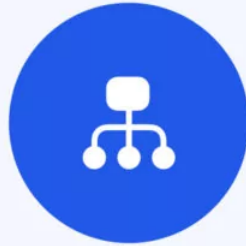


SCRUM ARTIFACTS



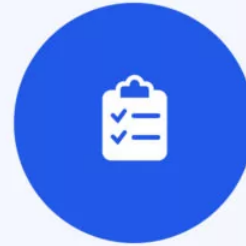
Product
backlog

+



Sprint
backlog

+



Product
increment



Shippable
software

Artifacts

SCRUM VALUES ... (1)



COURAGE

Scrum Team members have courage to do the right thing and work on tough problems



FOCUS

Everyone focuses on the work of the Sprint and the goals of the Scrum Team



COMMITMENT

People personally commit to achieving the goals of the Scrum Team



RESPECT

Scrum Team members respect each other to be capable, independent people



OPENNESS

The Scrum Team and its stakeholders agree to be open about all the work and the challenges with performing the work

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COMMITMENT

- ☐ The first value that underpins the proper implementation of scrum is the **commitment** to the work.
- ☐ A Scrum oriented Development Team has to be committed to Sprint Goal and Product Goal, but even more important is the commitment of each individual Developer.
- ☐ It provides not only an active effort to solve problems that arise in the course of daily duties. It also includes the willingness to support other members of the Development Team when they need it



FOCUS

- ❑ If each of the Developers is focused on sub-tasks with full commitment, it can be easy to lose sight of the main Product Goal and even the Sprint Goal.
- ❑ As a result, the Increment produced in a given Sprint will turn out to be incomplete.
- ❑ Therefore, focusing on Sprint Goal is extremely essential from a team perspective.
- ❑ Focus improves not only the level of tasks performed by individual Developers but also, their results come together to achieve Sprint Goal.
- ❑ And thus, the business value is realized: Scrum Team's workflow enables the creation and development of Product functionality



OPENNESS

- ❑ The agile approach differs from the traditional approach primarily in its openness to change.
- ❑ That is why it was among the main values of Scrum.
- ❑ Both the Scrum Team and the Stakeholders benefit by cultivating the value of openness in their work.
- ❑ The variability in question can refer to many areas of the project, for example: reduction of the project scope, redefinition of Product functionality, reorganization of tasks performed by Scrum Team to optimize efficiency.
- ❑ Changes can result from Stakeholder decisions, business or technical constraints.
- ❑ Openness in the face of change is supported by transparency, which is one of the pillars of empiricism in Scrum.
- ❑ Thanks to transparency, Scrum Team members and Stakeholders know what the change results from and where it can lead

**RESPECT**

- ☐ Scrum Team members are independent, equal individuals.
- ☐ Developers, Scrum Master and Product Owner are all professionals who treat each other with due respect. There should be no room for hierarchy in a Scrum Team.
- ☐ Respect also serves to provide psychological well-being.
- ☐ This is especially true for members of the Development Team.
- ☐ By showing respect to each other, they can experiment freely and not be afraid to make mistakes, which are a necessary part of the innovation process and lead to improved Scrum Team performance.



COURAGE

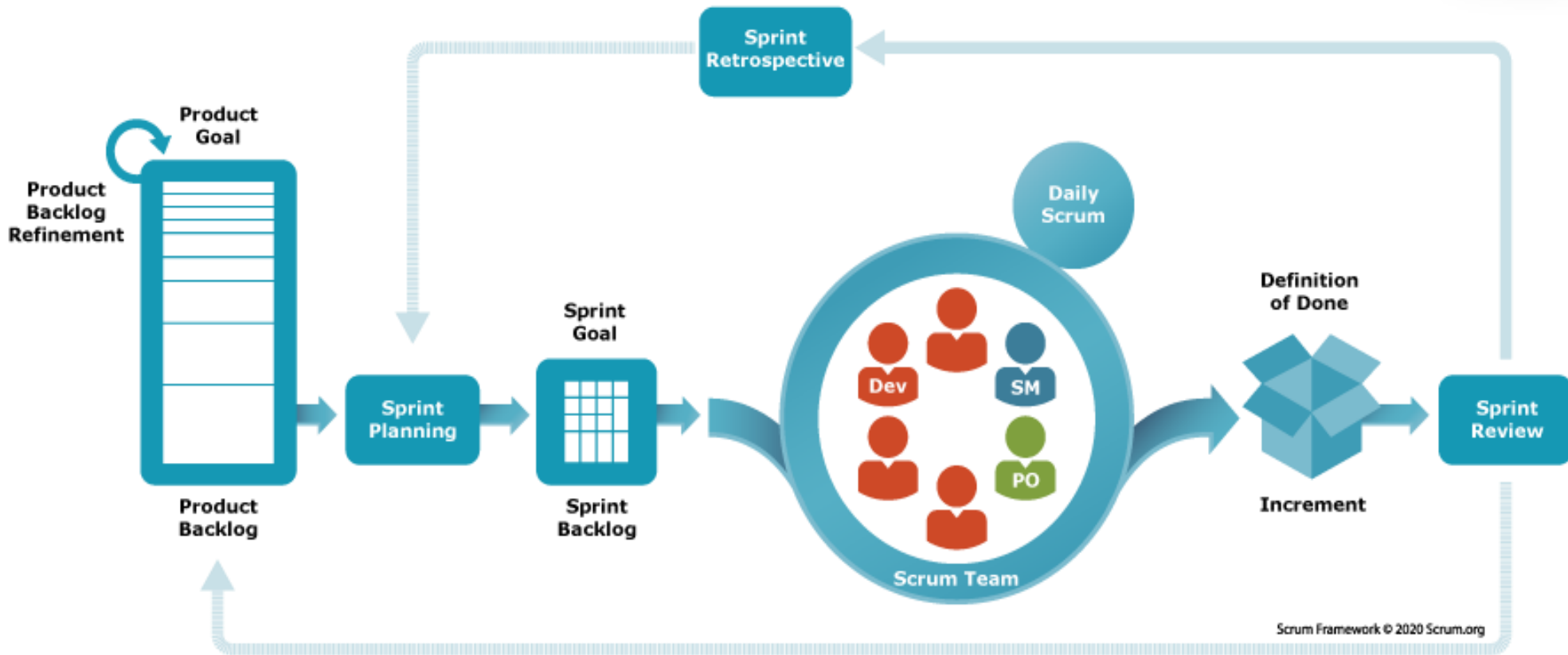
- ❑ Courage in problem-solving is, along with respect, a prerequisite for successful experimentation.
- ❑ It also makes the Scrum Team, when faced with difficult tasks, repeat attempts to solve them, until obtaining a satisfactory result.
- ❑ Being courageous in proposing business, technical, and interpersonal solutions helps to realize the innovation potential inherent in Scrum.
- ❑ Courage is also the foundation of self-management of the Scrum Team, which takes full responsibility for the consequences of the decisions made by the team members.



- ❑ The scrum framework is a structured approach to agile project management methodology.
- ❑ It consists of several components, including:
 - **Sprints**
 - **Product backlog**
 - **Sprint backlog**
 - **Scrum artifacts**
 - **Scrum events**
- ❑ The scrum framework emphasizes flexibility, adaptability, and continuous improvement, making it a popular choice for managing complex projects in an agile manner.



SCRUM FRAMEWORK





SPRINTS

Sprints_ are time-boxed iterations, typically two-to-four weeks in length, where the development team works to complete a set of planned work items.

PRODUCT BACKLOG

The product backlog is a prioritized list of all the features, enhancements, and bug fixes the product needs to address. The product owner manages and maintains this backlog.

SPRINT BACKLOG:

The sprint backlog is a subset of the product backlog, containing the work items selected for a specific sprint.



SCRUM ARTIFACTS

Scrum employs various artifacts, such as burn-down charts and task boards, to track progress and make the work visible to the team and stakeholders.

SCRUM EVENTS

Scrum events, including sprint planning, daily stand-ups, sprint reviews, and sprint retrospectives, provide structure and opportunities for collaboration within the team.



TEACHING YOU IS GOOD LUCK